

STUDENT PROJECT DOCUMENTATION

EduGenie

Google Gemini Powered Learning Assistant

A lightweight AI-powered educational assistant for question answering, concept explanation, summarization, quiz generation, and personalized learning recommendations.

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Program: Bachelor of Computer Applications (BCA)

Institution: St Thomas College of Arts and Science
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Project Team Details

This documentation records the design, implementation, and functional demonstration of EduGenie, a Google Gemini powered learning assistant developed as a student project in the Bachelor of Computer Applications programme.

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Declaration

The team presents this document as the project documentation for EduGenie. The contents describe the application workflow, module design, backend API structure, frontend interface, and functional testing demonstrations included with the project.

Submitted by

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Abstract

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. The system allows students to ask questions, understand difficult concepts through simplified explanations, summarize educational content, generate quizzes, and receive personalized learning recommendations.

The application uses a FastAPI backend with a simple HTML and CSS frontend. Cloud-based Gemini models are accessed through API calls, allowing the application to provide intelligent responses without requiring heavy local computing resources. The project also includes API-key pooling and error handling to manage rate limits more gracefully.

Key Features

- Question answering for general knowledge and academic queries.
- Step-by-step explanations of complex topics in student-friendly language.
- Automatic summarization of long educational passages.
- Generation of multiple-choice quizzes with answer validation.
- Structured learning recommendations from beginner to advanced levels.
- API-key pooling to reduce disruption caused by temporary rate limits.

1. Project Description

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. Designed for students of different academic levels, it provides an interactive environment where learners can enter a topic, question, or paragraph and receive useful educational output.

Purpose of the Project

The purpose of EduGenie is to make common learning tasks faster and easier by bringing multiple study-support functions into one web interface. Instead of using separate tools for searching, explaining, summarizing, and quiz preparation, the user can access these functions from the same application.

Core Functions

- Ask questions and receive concise answers.
- Understand complex concepts through simplified explanations.
- Generate quizzes from a topic or supplied content.
- Receive structured learning recommendations.
- Summarize large educational passages into readable notes.

Scenarios

- **Scenario 1:** A student wants to know about oceans and rivers and asks, "Which is the largest ocean?".
- **Scenario 2:** A student wants to check her understanding of the Pythagoras Theorem and uses the quiz-generation function.
- **Scenario 3:** A learner exploring SQL requests a structured learning path covering beginner, intermediate, and advanced topics.

2. System Architecture and Workflow

Architectural Flow

The application follows a simple request-response architecture. The student interacts with the frontend, the FastAPI backend routes the request, and the selected module calls the Gemini service to generate an appropriate response.

- **Start:** The user opens the EduGenie frontend and provides an input.
- **FastAPI Backend:** The user selection is mapped to the appropriate API endpoint.
- **Module Processing:** Requests are handled by the Explanation, Q&A, Quiz, Summarization, or Learning Path module.
- **End:** The generated result is returned to the frontend and displayed to the user.

Backend API Modules

Function	Endpoint / Module	Purpose
Explanation	/explain/	Provides simplified educational explanations.
Q&A	/qa	Answers direct questions and general knowledge queries.
Quiz	/quiz	Generates multiple-choice questions with four options.
Summary	/summarize/	Condenses long text into a shorter readable summary.
Learning Path	/learn/recommendations	Builds structured recommendations for learning a topic.

Technology Stack

- Python 3.10 or later.
- FastAPI framework with Uvicorn ASGI server.
- HTML and CSS for the frontend, with Jinja2 templating.
- Google Gemini API access through Python SDKs.
- Environment variables for securely storing API keys.

3. Project Architecture and Prerequisites

Folder Architecture

```
EduGenie/
|-- main.py           # FastAPI app
|-- explanation_module.py # Concept explanation logic
|-- qna.py            # Question answering
|-- quiz_module.py    # Quiz generation
|-- summary_module.py  # Summarization
|-- learning_path.py   # Learning recommendations
|-- templates/index.html # HTML frontend
|-- static/style.css   # Styling
`-- requirements.txt   # Python dependencies
```

Pre-requisites

- **Python 3.10+:** Required to run the application and its dependencies.
- **FastAPI and Uvicorn:** Used to expose and serve the backend API.
- **HTML, CSS, and Jinja2:** Used to create and render the web interface.
- **Gemini API keys:** Multiple keys can be configured as environment variables such as GEMINI_API_KEY_1, GEMINI_API_KEY_2, and GEMINI_API_KEY_3.

Model Selection

Gemini 3.8 Flash: Used for Q&A, explanations, summarization, and quiz generation because these functions benefit from fast responses.

Gemini 3.8 Pro: Used for more structured learning-path generation and deeper reasoning tasks.

4. Development Methodology

Milestone 1 - Model Selection and Architecture

The initial milestone focused on selecting AI models, identifying the main application modules, and defining a modular folder structure. The design separates each educational capability into an individual Python module so that features can be maintained and improved independently.

Milestone 2 - Core Functionalities Development

Explanation Module: Uses a fast Gemini model to convert a topic into clear, educational breakdowns. The implementation includes a `get_random_key()` function to select from available API keys and an error-handling path for temporary rate-limit conditions.

QnA Module: Accepts a user question, authenticates with the selected API key, sends the question to Gemini, and returns the generated text to the frontend.

Quiz Module: Instructs the model to behave as a quiz generator. It requests valid JSON containing three multiple-choice questions, four options per question, and one correct answer. A `clean_json_block` function removes Markdown code fences before parsing the response.

Summary Module: Accepts large text blocks and generates simple summaries. It also uses protected API calling and exception handling to reduce application failures during rate limiting.

Learning Path Module: Generates structured recommendations for a topic. The prompt asks for key topics, learning order, and resources across beginner, intermediate, and advanced levels.

Milestone 3 - Frontend Development

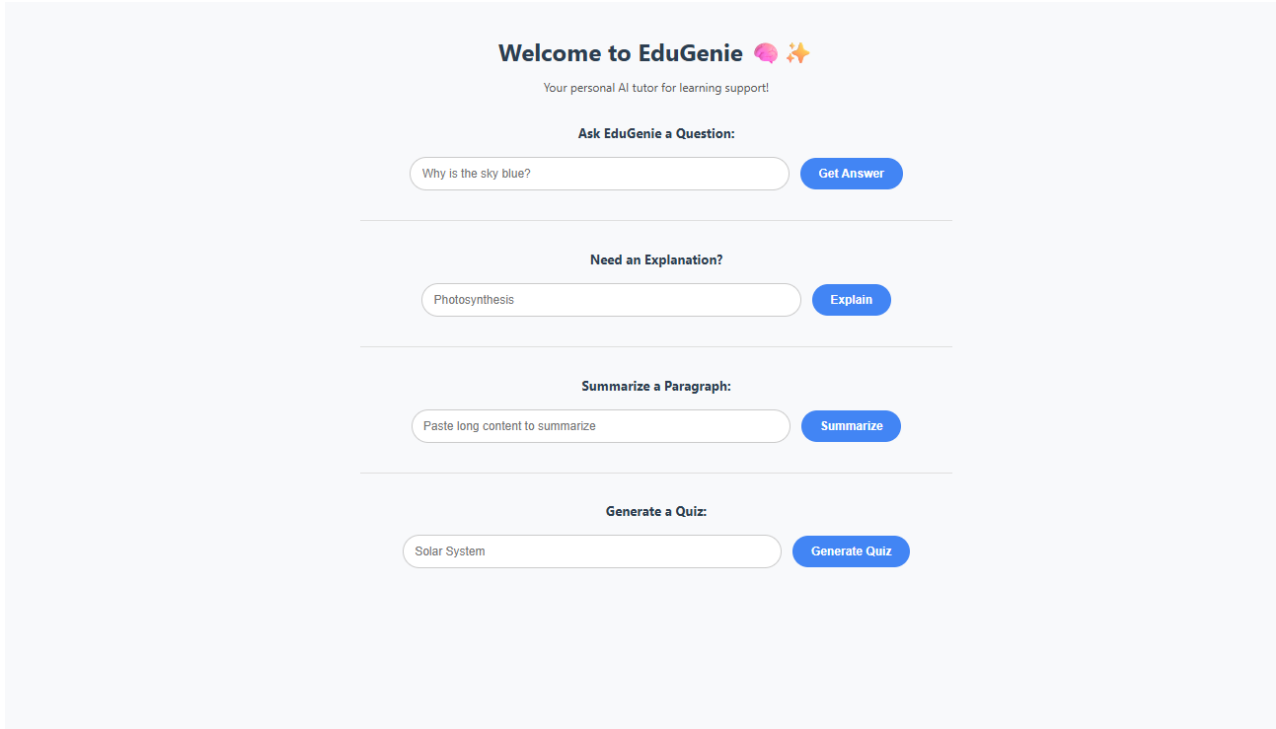
The frontend is served using Jinja2 templates and provides a clean browser-based interface for the main functions. The user can enter a question, a topic for explanation, text for summarization, or a topic for quiz generation.

Milestone 4 - Functional Testing and Demonstration

The final milestone focused on running the application and validating that the main features produce visible results. The screenshots in the next section document the interface and example outputs produced during testing.

5. Frontend and Functional Demonstration

The following screenshots were captured during project testing. They are included as evidence of the working user interface and the generated outputs from the Q&A, explanation, summarization, and quiz features.



The screenshot displays the EduGenie home page with a light blue background. At the top, it says "Welcome to EduGenie" with a pink brain icon and a yellow star icon, followed by the tagline "Your personal AI tutor for learning support!". Below this, there are four distinct sections, each with a title, a text input field, and a blue action button. The first section is "Ask EduGenie a Question:" with the input "Why is the sky blue?" and a "Get Answer" button. The second is "Need an Explanation?" with the input "Photosynthesis" and an "Explain" button. The third is "Summarize a Paragraph:" with the input "Paste long content to summarize" and a "Summarize" button. The fourth is "Generate a Quiz:" with the input "Solar System" and a "Generate Quiz" button. Each section is separated by a thin horizontal line.

Figure 1. EduGenie home page and available learning features.

Welcome to EduGenie

Your personal AI tutor for learning support!

Ask EduGenie a Question:

why is ocean i blue

Get Answer

The ocean is blue primarily because of **how water absorbs and scatters sunlight**.

Here is the breakdown of why this happens:

1. Water Absorbs Colors of Light

Sunlight looks white to us, but it is actually made up of all the colors of the rainbow combined (red, orange, yellow, green, blue, indigo, and violet). Each color travels at a different wavelength:

- Red, orange, and yellow** have long wavelengths.
- Blue and violet** have short wavelengths.

When sunlight hits the ocean, water molecules act like a filter. Water is very good at **absorbing longer wavelengths**. It quickly absorbs red, orange, and yellow light near the surface and turns that light energy into heat.

2. Blue Light Travels Deeper and Scatters

Because blue light has a short wavelength, it passes through the water much more easily and travels deeper. Eventually, this blue light collides with water molecules and **scatters** in every direction. Some of that scattered blue light bounces back up out of the water and into your eyes, making the ocean look blue.

Common Questions & Myths:

- Isn't the ocean blue because it reflects the sky?**
- Mostly a myth! While the surface of the water **can** act like a mirror and reflect a blue sky (or gray clouds), the ocean is blue all on its own. Even inside an indoor, deep swimming pool with white walls, the water looks blue.
- Why is a glass of water clear?**
- There simply aren't enough water molecules in a single glass to absorb enough red light. You need a massive amount of water (like a deep pool or an ocean) for the blue color to become visible.
- Why are some parts of the ocean green or brown?**
- Green:** Often caused by microscopic marine plants called **phytoplankton**, which contain green chlorophyll to absorb sunlight.
- Brown or Murky:** Caused by sand, mud, and sediment stirred up by waves near coastlines or washed in by rivers.

Figure 2. Question-answering response for an ocean-related query.

Welcome to EduGenie

Your personal AI tutor for learning support!

Ask EduGenie a Question:

why rose is red ?

Get Answer

A rose is red primarily due to a combination of **chemistry, biology, and human intervention**.

Here is a breakdown of why roses are red:

1. The Chemical Cause: **Anthocyanins**

The red color comes from a group of natural plant pigments called **anthocyanins** (specifically one called **cyanidin**).

* These pigments are stored inside the cells of the rose petals.

* Anthocyanins absorb blue, green, and ultraviolet light from the sun, but **reflect red light**. When this reflected light reaches your eyes, your brain sees the petals as red.

* The exact shade of red also depends on the **pH level** inside the petal's cells; higher acidity usually produces a brighter, more vibrant red.

2. The Evolutionary Purpose: **Pollination and Protection**

In the wild, plants do not produce colors just to look pretty; they do it to survive:

* **Attracting Pollinators:** Bright petals stand out against green leaves, helping birds, butterflies, and insects find the flower easily so they can pollinate it.

* **Sun Protection:** Anthocyanins act as a natural sunscreen for the plant. They protect the delicate petals from damage caused by harmful UV rays and free radicals.

.....

Figure 3. Question-answering response explaining why roses are red.

3. The Human Role: **Selective Breeding**

Most wild roses are actually pale pink, white, or yellow—deep red wild roses are quite rare.

The vibrant red roses we see today exist mostly because **humans selectively bred them**. For thousands of years (especially starting in China and later in Europe), gardeners crossed different rose species together to make the red color richer, darker, and longer-lasting because red became a universal symbol of love, beauty, and passion.

Need an Explanation?

how sea turtle live ?

Explain

Sea turtles are ancient reptiles that have lived in Earth's oceans for over 100 million years—meaning they once shared the planet with dinosaurs. Although they spend almost their entire lives in the sea, they are air-breathing reptiles with a fascinating and complex life cycle.

Here is a clear, educational overview of how sea turtles live, from birth to adulthood.

1. How Their Life Begins: The Nesting Cycle

A sea turtle's life begins on land, even though they spend the rest of their days in the water.

* **Laying the Eggs:** Female turtles drag themselves onto sandy beaches at night. They dig a hole with their rear flippers, lay roughly **80 to 120 soft-shelled eggs**, cover the nest with sand to protect it from predators, and immediately return to the sea.

* **Incubation:** The eggs stay buried for about 60 days. The temperature of the sand determines the sex of the turtles: **warmer sand produces females, while cooler sand produces males.**

* **The Scramble to the Sea:** The babies hatch underground, dig their way out together at night, and rush toward the ocean, guided by the natural light of the moon reflecting on the water. This is the most dangerous time of their lives, as crabs, raccoons, and birds wait to eat them.

2. The "Lost Years" (Early Life)

Figure 4. Explanation module response for the sea turtle life cycle.

2. The "Lost Years" (Early Life)

Once the hatchlings reach the ocean, they enter a period scientists long called the **"Lost Years."**

*** The tiny turtles swim into open ocean currents, traveling thousands of miles.**

*** They spend several years drifting in large floating mats of seaweed (such as *Sargassum*). This seaweed provides them with food (tiny shrimp and fish) and shelter from large predators.**

3. Underwater Survival: How Their Bodies Work

Sea turtles are uniquely built for ocean life:

*** *Breathing:*** Like humans, sea turtles have lungs and must come to the surface to breathe air. However, while resting or sleeping, they can slow their heart rate and hold their breath for **4 to 7 hours** underwater!

*** *Swimming:*** Unlike land turtles with claws, sea turtles have long, paddle-like front flippers for swimming and shorter rear flippers used as rudders for steering.

*** *Drinking Salt Water:*** Because they live in the ocean, they have special glands near their eyes that "pump" excess salt out of their bodies. When they come ashore, this looks like they are crying, but they are simply keeping their bodies balanced.

*** *Navigation:*** Sea turtles have a built-in magnetic "GPS." They can detect the Earth's magnetic fields, allowing them to navigate across vast oceans without getting lost.

4. What Do Sea Turtles Eat?

Different species of sea turtles have different diets, which keeps marine ecosystems healthy:

*** *Green Turtles:*** Mostly herbivores. They eat seagrass and algae, acting like underwater lawnmowers to keep seagrass beds healthy.

*** *Leatherback Turtles:*** The largest turtles, they eat almost exclusively **jellyfish**. They have backward-pointing spikes inside their throats to keep slippery jellyfish from floating back out.

*** *Loggerhead Turtles:*** Have massive jaws designed to crush hard-shelled prey like crabs, conchs, and clams.

*** *Hawksbill Turtles:*** Have narrow, bird-like beaks to reach into coral reefs and eat sea sponges.

5. Growing Up and Returning Home (Natal Homing)

Sea turtles grow very slowly. It often takes **20 to 30 years** for a sea turtle to reach adulthood and reproduce.

Figure 5. Continued sea turtle explanation showing early life and survival.

6. Threats to Sea Turtle Life

In nature, sea turtles can live for **50 to 100 years**, but human activity makes survival difficult:

* **Plastic Pollution:** Turtles often mistake floating plastic bags for jellyfish, which can starve or choke them.

* **Fishing Nets (Bycatch):** Turtles can become trapped in commercial fishing gear and drown because they cannot reach the surface to breathe.

* **Artificial Lighting:** Lights from beachfront hotels and streets confuse baby turtles, leading them away from the ocean toward roads and predators.

Summary

Sea turtles live as nomadic wanderers of the ocean. From their dramatic sprint to the water as hatchlings, to traveling across whole oceans, and finally returning to their birth beach decades later, their lives are a remarkable journey of endurance and survival.

Summarize a Paragraph:

A sea turtle's life begins on land, even though they spend the rest of

Summarize

Summary:

Sea turtles start their lives on land. At night, a mother turtle digs a hole on a sandy beach, lays about 80 to 120 eggs, buries them for safety, and returns to the ocean.

About two months later, the eggs hatch. The warmth of the sand decides their sex: warmer sand makes females, while cooler sand makes males. Finally, the baby turtles dig out of the nest together at night and race to the water using the moonlight, trying to avoid predators like birds and crabs along the way.

Figure 6. Continued explanation and generated summary of sea turtle nesting.

Generate a Quiz:

sea creatures

Generate Quiz

Quiz:

Q1: Which of the following is the largest living sea creature on Earth?

☐ Great white shark

☒ Blue whale

☐ Giant squid

☐ Walrus

Check Answer

☒ Correct!

Q2: Which sea creature is known for having three hearts and blue blood?

☒ Octopus

☐ Sea turtle

☐ Jellyfish

☐ Dolphin

Check Answer

Figure 7. Quiz generation interface with multiple-choice questions.

Q2: Which sea creature is known for having three hearts and blue blood?

- ☒ Octopus
- ☐ Sea turtle
- ☐ Jellyfish
- ☐ Dolphin

Check Answer

✓ **Correct!**

Q3: In which of these sea creatures does the male carry and give birth to the young?

- ☐ Clownfish
- ☒ Seahorse
- ☐ Starfish
- ☐ Moray eel

Check Answer

✓ **Correct!**

Figure 8. Quiz answer validation showing correct responses.

6. Testing Summary and Observations

1. Asking Questions

The Q&A functionality accepted natural-language questions and displayed generated educational explanations. Example prompts included questions about the ocean and the color of flowers.

2. Explanations

The explanation feature produced structured, student-friendly responses. The sea turtle example demonstrated how a topic can be broken into life-cycle stages, survival features, diet, growth, and threats.

3. Summarizing Paragraphs

The summarization feature condensed a longer passage about sea turtle nesting into a shorter explanation while retaining the main ideas.

4. Generating Quizzes

The quiz module generated multiple-choice questions and displayed answer-validation feedback. The sea-creatures demonstration shows the correct-answer state for several questions.

Challenges and Handling

One of the main development challenges was handling API rate limits. The project addresses this by selecting from multiple configured API keys and returning a user-friendly cooldown message when the available keys are temporarily unavailable. This approach helps prevent the application from terminating unexpectedly when a request cannot be served immediately.

7. Conclusion

EduGenie has been developed as an accessible AI-powered educational assistant that combines cloud-based generative AI with a lightweight web interface. The modular structure supports question answering, concept explanation, summarization, quiz generation, and learning recommendations within one application.

The project demonstrates how a FastAPI backend can be integrated with Gemini API services to build practical educational software. The inclusion of API-key pooling and exception handling also provides a more resilient user experience under temporary rate-limit conditions.

EduGenie provides a foundation for further enhancements such as richer quiz analytics, user progress tracking, more personalized study plans, authentication, persistent learning history, and additional educational modules.

Project Team

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References

- FastAPI documentation - framework and API development reference.
- Uvicorn documentation - ASGI server reference.
- Jinja2 documentation - HTML templating reference.
- Google Gemini API / Google GenAI developer documentation - model integration reference.

End of Student Project Documentation